Kaya Han Taş

Data Scientist | Astrophysicist | Python Developer
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PROFILE

Analytical and detail-oriented MSc graduate in Physics & Astronomy (cum laude) with experience analyzing complex datasets, developing reproducible Python workflows, and translating technical analyses into clear insights. Passionate about applying data analytics and machine learning to real-world problems, with a strong interest in growing as a data scientist in a collaborative, impact-driven environment.

EDUCATION

University of Amsterdam & Vrije Universiteit Amsterdam **Sep. 2023 – Aug. 2025**
Master of Science in Physics & Astronomy *Amsterdam, NL*

- **GPA: 8.6/10.0** (Dutch System); *Cum Laude*
- Completed a research-focused MSc with a strong focus on data analysis, statistical methods, and time-series modelling.
- Conducted an end-to-end data analysis project leading to a peer-reviewed publication (first-author).
- Developed a Python-based data analysis tool called *ExoMUSE* and applied it to real-world datasets.

Akdeniz University **Sep. 2018 – Jun. 2022**
Bachelor of Science in Space Sciences and Technologies *Antalya, TR*

- **GPA: 3.89/4.00** (US System); *Summa Cum Laude, Top student in the Faculty of Science*
- Designed and developed an interactive Python GUI for modelling time-series datasets.
- Conducted research in detection pipelines and remote sensing.

PROJECTS & RESEARCH EXPERIENCE

TOI-2431 b - Planet Discovery with Time-Series Modelling

- Built Python data pipelines to clean, integrate, and analyze multi-source time-series data.
- Applied Bayesian inference and probabilistic modelling to extract weak signals from noisy datasets and support reliable, data-driven conclusions.
- Evaluated and optimized model performance using ExoMUSE and other open-source Python packages.
- Validated discovery of a new exoplanet with high-confidence (>5σ); demonstrated ability to handle large, complex datasets (~6000 planets, each with over 20 parameters) and collaborated with international teams.
- Published in *Astronomy & Astrophysics* (2026), DOI: 10.1051/0004-6361/202556365

ExoMUSE – Python Package for Data Analysis

- Developed a modular Python package for parameter estimation and uncertainty quantification.
- Implemented sampling-based optimization methods for high-dimensional, noisy datasets.
- Enabled joint modelling of multiple dataset types to improve result accuracy and robustness.
- Applied to real-world data to validate performance and improve visualization and accessibility.
- Available on GitHub: https://github.com/Kaya-Han-Tas/ExoMUSE.

Facial Emotion Recognition – Machine Learning Project

- Built and trained a convolutional neural network (CNN) in PyTorch to classify facial expressions.
- Improved model accuracy to 62% using cross-entropy loss and Adam optimizer.
- Applied data preprocessing, model evaluation, and ML workflows.
- Jupyter notebook available on GitHub.

Light Curve Modelling – Time-series data analysis tool with GUI

- Developed a Python-based tool to model time-series data using regression and validation workflows.
- Produced clear visual outputs and automated LaTeX reporting to enable effective analysis and communication of results.
- Available on GitHub: https://github.com/Kaya-Han-Tas/Light-Curve-Modelling.

TECHNICAL SKILLS

- **Programming & Tools:** Python (NumPy, Pandas, Matplotlib, Seaborn, SciPy, scikit-learn, PyTorch), SQL, AWS Cloud Computing, Jupyter, Git, Linux
- **Data Science & Analytics:** Time-Series Analysis, Bayesian Inference, Regression, Machine Learning (Supervised & Deep Learning), MCMC, Monte Carlo Simulation, Signal Processing, Statistics
- **Other:** QGIS/ArcGIS, AutoCAD, Remote Sensing, Photogrammetry, GUI design (Tkinter), LaTeX

CERTIFICATIONS & AWARDS

- KHMW National Prize for Best MSc Thesis in Astronomy *(2025)*
- API Best MSc Thesis Prize, University of Amsterdam *(2025)*
- Observation Proposal Granted – Mercator Telescope *(2024)*
- Certificate of High Honor, Akdeniz University *(2022)*
- Top Student of Faculty of Science, Akdeniz University *(2022)*
- Top Student of Space Sciences and Technologies, Akdeniz University *(2022)*
- ERASMUS+ "Interculture Dedicated Education and Learning (IDEAL)", Certificate of Attendance *(Portugal, Hungary, Turkey, 2017)*

WORK EXPERIENCE

Amazon **Feb. 2026 – Current**
AI Content Expert II *Den Haag, NL*

- Curate, annotate, and validate high-quality datasets supporting the training and evaluation of AI models, including large language models (LLMs).
- Apply analytical judgement to evaluate complex, multimodal data and resolve ambiguous cases while maintaining high data quality standards.
- Collaborate with cross-functional teams to improve data quality, annotation workflows and AI model evaluation processes.

Albert Heijn **Oct. 2024 – Jan. 2026**
Stock Associate *Amsterdam, NL*

- Balanced MSc with part-time employment, demonstrating adaptability, teamwork, and time management.

ADDITIONAL INFORMATION

- **Languages:** English (Fluent), Turkish (Native), German (Basic), Dutch (Basic)
- **Soft Skills:** Analytical and Critical Thinking, Team Collaboration, Adaptability, Curiosity, Problem Solving, Communication
- **Interests:** Kickboxing, Music, Photography, Travelling, Gaming, Reading